

A Review of Current Trends in Greenhouse Gas Emission Prediction through Machine Intelligence Learning Techniques and Future Challenges

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Abstract—Increased GHG emissions from human activities have severe environmental and health impacts, making accurate predictions essential for effective mitigation strategies. This review explores various models for GHG emission prediction, focusing on ML DL techniques. It examines the contributions of sectors such as transportation, agriculture, and energy production, and discusses challenges in estimating emissions from sources like soils, livestock, and hydroelectric power. Prediction accuracy has significantly increased thanks to developments in ML and DL algorithms, such as RNNs, CNNs, and artificial neural networks (ANNs). Case studies highlight the effectiveness of these models in capturing complex emission patterns and informing emission reduction decisions. Integrating ML and DL with traditional environmental data analysis offers a comprehensive approach to GHG emission prediction, supporting sustainable development and climate change mitigation. Despite significant progress, further research is needed to enhance model interpretability, data integration, and real-time prediction capabilities to address the dynamic nature of GHG emissions and their global impact.

Keywords—Greenhouse Gas Emission, Emission Trends, Predictive Modeling

I. INTRODUCTION

GHGs like ozone, nitrous oxide, chlorofluorocarbons, methane and carbon dioxide play a crucial role in absorbing infrared radiation. This process traps heat in the atmosphere, leading to global warming. The Earth's temperature has been on a steady rise over the last century [1]. Global warming results in heat-related health issues, like heat strokes, which can lead to premature deaths, especially during the summer months. Greenhouse gases (GHGs) function like a partial blanket, trapping heat by radiating long-wave radiation and absorbing warmth from the Earth's surface. Human activities, particularly since the Industrial Revolution, have significantly increased the emissions of major GHGs, such as CO₂, CH₄, N₂O, and O₃ [2]. Many of these gases persist in the atmosphere for a long time, which makes their impacts almost impossible to reverse. A significant challenge to achieving sustainable development and creating greener cities is the issue of greenhouse gas emissions [3]. Human activities, especially in transportation, play a major role in contributing to greenhouse gas emissions. In response, various alternative fuels and powertrains have been developed for road transportation to help reduce these emissions [4]. The expanding airline industry, characterized by crowded air routes, leads to delays, increased fuel consumption, and higher

greenhouse gas emissions [5]. The automobile industry also uses substantial quantities of gasoline and electricity [6]. Minimizing the environmental impact of transportation systems is essential [7]. Agriculture plays a significant role in greenhouse gas emissions. Changes in land use and farming practices, especially in livestock production, release methane [8]. Manure management on dairy farms, particularly the use of liquid manure, can lead to an increase in greenhouse gas emissions [9]. While hydroelectric power is often seen as a renewable energy source, it can still lead to greenhouse gas emissions due to the decay of vegetation and emissions associated with reservoirs [10]. Even after decades of research, predicting and reducing greenhouse gas emissions accurately continues to be a significant challenge in the realm of sustainable development.

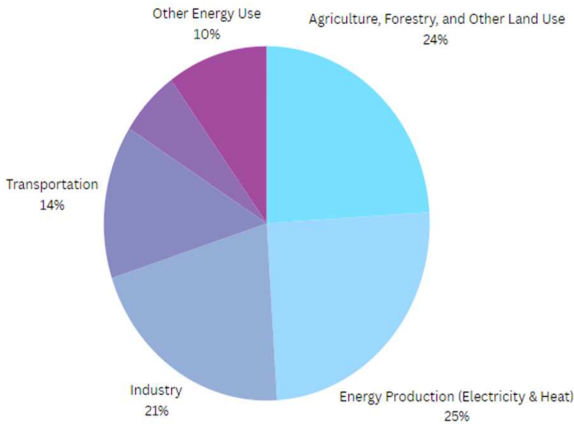


Fig. 1 Greenhouse Gas (GHG) Emission Sources

II. MACHINE LEARNING AND DEEP LEARNING GREENHOUSE GAS EMISSIONS PREDICTION

A. Greenhouse gas emission and effect

Global economies and the environment are seriously threatened by climate change. The most common greenhouse gas, carbon dioxide, is largely to blame for the one degree Celsius increase in the average world temperature during the past 150 years. [11]. The greenhouse effect, which occurs in the troposphere, maintains Earth's average temperature at

14°C rather than allowing it to drop to -19°C [12]. Climate change impacts plant life by increasing CO₂ levels, which boosts photosynthesis, but it also reduces the opening of stomata. Additionally, ozone (O₃) can create oxidative stress and impede the process of photosynthesis[13]. Predictions indicate that we can expect more severe weather events, including heatwaves, droughts, and floods, as well as changes in rainfall patterns[14]. Fossil fuels account for 85% of the world's energy production, which have a profound impact on ecosystems and human health [3,8,15]. By the year 2100, global temperatures could increase by 4 to 5°C, leading to a higher occurrence of extreme weather events around the world [15]. High levels of CO₂ can lead to health problems like shortness of breath, dizziness, and confusion when inhaled in significant amounts [16]. The agricultural sector is confronted with the challenge of lowering greenhouse gas (GHG) emissions while ensuring food security. Agriculture and changes in land use are responsible for approximately 25% of global GHG emissions, with rice paddies contributing 20% of methane emissions and 10% of nitrous oxide emissions [17,18]. Finding ways to reduce these emissions while still ensuring agricultural productivity is a critical challenge for sustainable development.

B. Harnessing AI: Deep Learning and Machine Learning for Predicting Green House Gas Emissions

Machine learning and deep learning, two subfields of artificial intelligence, have transformed a variety of industries by allowing computers to learn from data and make decisions on their own without explicit programming[19]. Machine learning algorithms recognize patterns in extensive datasets, whereas deep learning models emulate brain functions to handle intricate information [20]. Machine learning is divided into three basic categories: unsupervised, supervised, and reinforcement learning. Unsupervised learning focuses on discovering patterns in unlabeled data, whereas supervised learning uses labeled data to train the model. [21]. Training agents to make a series of decisions based on a reward system is Reinforcement learning. Deep learning emphasizes the use of artificial neural networks that consist of interconnected layers of data-processing nodes. It performs exceptionally well in areas such as NLP, image recognition, and autonomous driving [22, 23]. While recurrent neural networks (RNNs) are excellent at processing sequential input [24].

AI is crucial for estimating greenhouse gas (GHG) emissions, especially through machine learning and deep learning. It analyzes large datasets obtained from sources such as satellite images, climate models, and sensor networks [25]. These algorithms can pinpoint emission hotspots, monitor changes over time, and offer insights for focused emission reduction strategies [26]. By examining aspects like energy use, transportation, and land management, AI can assist researchers and policymakers in creating effective strategies for mitigation. Machine learning models can also forecast future emissions and identify early indicators of high-emission activities, such as deforestation or industrial outputs, which aids in enhancing efforts to reduce emissions[27, 28].

C. Unleashing the Power of Deep Learning and Machine Learning for Green House Gas Emissions

Machine learning (ML), which is a branch of AI, utilizes regression, classification, and prediction models that achieve

high accuracy in various industries thanks to its robust capabilities and effective training methods. As computational power continues to grow, statistical techniques are being more frequently employed to examine intricate environmental issues [29]. For example, machine learning models were used to assess nitrous oxide (N₂O) and carbon dioxide (CO₂) emissions over a five-year period in Quebec, Canada. In Turkey, various methods such as SVM, DL, and Artificial Neural Networks (ANN) were employed to forecast greenhouse gas emissions, with the DL model delivering particularly accurate outcomes [30].

III. NAVIGATING THE STORM : OVERCOMING CHALLENGES IN ML AND DL FOR GREEN HOUSE GAS PREDICTION

Accurate predictions of greenhouse gas (GHG) emissions are essential for tackling climate change, yet ML and DL models encounter various challenges. These models need substantial amounts of labeled data to reach high levels of accuracy, but obtaining this data for GHG forecasts can be both costly and time-consuming. Using previously trained models from related jobs is known as transfer learning, although it might not adjust effectively to data unique to greenhouse gas emissions. GHG emissions are affected by a variety of factors, it can be challenging to identify relevant features from complex datasets, and this often requires specialized knowledge, particularly when accounting for time-dependent elements like seasonal variations and shifts in policy. Models that rely on historical data might find it challenging to forecast emissions in new situations because of changes in technology, regulations, or socioeconomic factors. This highlights the need for ongoing updates to ensure accurate predictions. Deep neural networks are great at spotting intricate patterns, but their "black-box" nature brings up issues regarding interpretability, particularly in important decision-making scenarios. Finding a balance between model complexity and interpretability continues to be a challenge in greenhouse gas predictions. Moreover, issues with data availability and quality, like missing values or inconsistencies, can hinder the performance of machine learning and deep learning models, making it even more challenging to forecast greenhouse gas emissions accurately.

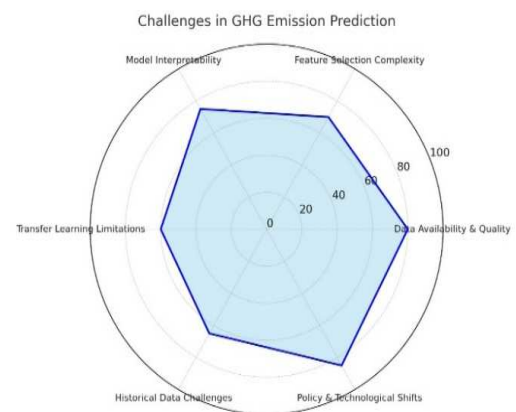


Fig.2 Radar chart visualizing the challenges in greenhouse gas (GHG) emission prediction.

In the Beijing-Tianjin-Hebei, China, a significant area for coal-based energy use, Menglu et al. examined the primary energy sources that contribute to carbon emissions, such as coal, gasoline, and natural gas. By analyzing energy consumption and carbon emission data from 2000 to 2016, they found that the SVM-ELM model surpassed the individual SVM and ELM models, achieving an RMSE of 12.3356% and an MRE of 1.623347% [29].

Wei Sun and associates created a model for predicting CO₂ emissions that combines particle swarm optimization for least squares support vector machine and principal component analysis (PCA). Their research, carried out in Hebei province, China, from 1990 to 2016, utilized PCA to reduce 24 important factors into two principal components. The PSO-LSSVM model achieved a prediction error of only 0.663%, surpassing traditional models like BPNN and LSSVM [25].

Igor and colleagues employed the SVM-FFA method to estimate CO₂ emissions about population growth within the European Union. By analyzing data from the World Bank, they discovered that the SVM-FFA model yielded highly accurate predictions, achieving RMSE and R-squared values of 4.2260 and 0.9015, respectively, based on gaseous fuel consumption [13].

A. Exploring Model Diversity: Analyzing Techniques for Green House Gas Emissions Prediction

Emissions of carbon dioxide (CO₂) are one of the most pressing environmental problems facing the world today since they contribute significantly to climate change. This study introduces a model designed to predict CO₂ emissions by utilizing regression tree and support vector machine (SVM) algorithms. The analysis was conducted using a dataset from Canada, covering the years 2014 to 2020. The results indicated that the SVM algorithm outperformed the others, achieving a performance of 3.6026% RMSE, 12.978% MSE, 2.57% MAE, and a perfect 100% R-squared, with a training duration exceeding 320 seconds [24].

Table 1. Summary of the existing studies using SVM classifier

References	Pollutant target	Accuracy
[7]	CO ₂ , CH ₄ , N ₂ O, F-gases, and total GHG emissions	R ² -0.926, RMSE- 8.858, MBE -1.69
[9]	Carbon dioxide (CO ₂) and nitrous oxide (N ₂ O) fluxes	R ² - 68% (prediction stage), R ² - 92%
[2]	Carbon dioxide (CO ₂)	C -0.1 and Epsilon - 0, Root Mean Square Error (RMSE) -0.004
[31]		MAPE -0.168%, MaxAPE-0.437%, MdAPE-0.117%
[10]	Green House Gases (GHG)	The maximum allowable error - 1.9
[62]	Carbon emission	MAPE - 1.61
[3]		mean absolute percentage error (MAPE)<0.03%
[8]	Carbon dioxide (CO ₂) emission	MAPE-1.1165%, RMSE-0.7013 and MAE-0.5425
[5]		RMSE -17.82, MAE-13.23

B. Summary of state-of-the-art related to Greenhouse Gas Emissions Prediction using SARIMA and ARIMA

Surbhi et al. used univariate time series data with three statistical models. The best performance was observed with LSTM, SARIMAX, and Holt-Winter, with SARIMAX achieving MAPE and RMSE values of 6.554 and 68.221, respectively [11]. Aishah et al. used statistical and ML models, such as LSTM, SARIMA, and ARIMA, to forecast NO₂ levels at 14 ground stations throughout the United Arab Emirates. The MAPE values for SARIMA were recorded at 22.66 for the open-loop scenario and 27.83 for the closed-loop scenario [12]. January 1 to June 8, 2016, utilizing a hybrid ARIMA-GRNN model. For the dataset without anomalous values (NO₂New), the coefficients of the ARIMA model were AR1 = 0.4671 and MA1 = 0.9083. The model produced an AIC of 15.95, MAE of 3.0694, MAPE of 30.76%, and an AIC of 770.53, with a residual variance of 15.8890 [42]. Amin et al. analyzed GOSAT data to examine the distribution of CO₂ across Iran from May 2009 to June 2016. Their SARIMA model produced an RMSE of 0.8 and a Pearson correlation of 0.93 for area 5 [43]. Shafiq et al. predicted methane levels in Pakistan, China, and India by employing ARIMA and LSTAR models. The ARIMA (2, 2, 0) model yielded the most accurate results for Pakistan and China, whereas the LSTAR model was most effective for India, achieving an RMSE of 192.78 and a ME of 168.94 [46].

Table 2. State-of-art works on Green House Gases using ARIMA and SARIMA

References	Pollutant Target	Method
[6]	Green House Gases (GHG)	SARIMA
[37]	Carbon Dioxide (CO ₂)	ARIMA and SARIMA
[35]	Carbon emission	SARIMA
[36]	PM10, PM2.5, SO ₂ , CO, and NO ₂	ARIMA and SARIMA
[20]	Carbon dioxide (CO ₂)	SARIMA
[21]		ARIMA and SARIMA
[41]	SO ₂ , NO _x , CO	ARIMA and SARIMA
[4]	Green House Gases	ARIMA
[4]	Green House Gases	ARIMA
[6]	Green House Gases (GHG)	SARIMA

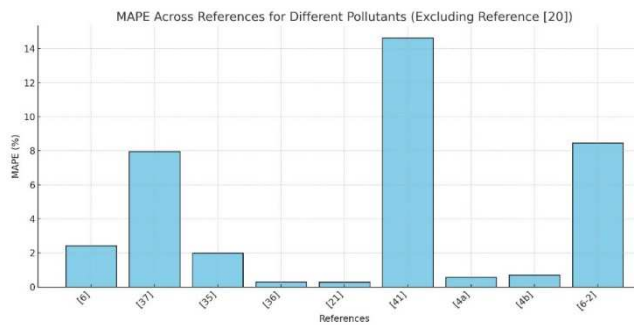


Fig.3 Mean Absolute Percentage Error (MAPE) across different references and pollutants.

C. Summary of state of the art work related to Greenhouse Gas Emissions Prediction using Long short-term memory network (LSTM)

Xianfu et al. sought to effectively monitor and estimate carbon emissions using the LSTM-trained SVR model demonstrated strong predictive capabilities, especially in forecasting elements such as carbon emissions and labor productivity in the "Yangtze River Economic Belt" basin [26]. A case study conducted by Jun and colleagues in Anhui, China, demonstrated that this method achieved a 35.21% reduction in RMSE for three pollutants, resulting in an RMSE of 8.15 [45]. Chan et al. created a regression model aimed at predicting carbon emissions from coal-fired power plants in China. By analyzing production logs from 2019 to 2021, the model produced MAE, RMSE, MAPE, and R2 values of 289.14, 357.15, 0.021, and 0.842, respectively [47]. Nur et al. predicted GHG emissions from manure management in Indonesia by employing LSTM and GRU models, analyzing data from 1961 to 2021. The models produced RMSE values of 4.25% for cattle, 3.61% for non-dairy cattle, and 3.25% for polluting birds [48].

IV. CONCLUSION

After Understanding climate change, assessing climate models, and helping to limit forecasts of anthropogenic climate change all depend on the detection of climate change and the identification of its causes. In its most recent report, the Intergovernmental Panel on climatic Change (IPCC) makes it abundantly evident that recent global warming is considerable in the context of natural climatic oscillations and that human activities are almost certainly to blame for this climate change. As a result, organizations, decision-makers, and the general public are looking to climate scientists for guidance on what they should do to mitigate dangerous climate change over the long term and adapt to the inevitable further climate change that will occur over the coming decades. To improve the precision and application of ML and DL models, however, a number of concerns, including as data availability and quality, feature selection and engineering, model complexity and interpretability, limited training data, and generalization problems, must be resolved. Interdisciplinary cooperation, standardized data gathering procedures, and additional research on model interpretability and generalization methods are necessary to address these issues.

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